

4/7

Phase II Project Demo

Cancer Pharmacogenomics

Business Understanding

The Problem:

During cancer treatment patients often have adverse side effects to different cancer drugs, however, these reactions can be reduced with more effective prescribing. Genetic mutations and drug responses are dependent on one another therefore understanding the sensitivity of cancer cell lines to different drugs can improve individuals cancer treatment experiences overall.

Our mission is to predict the sensitivity of cancer cell lines to cancer drugs using cells biological and pharmacological profile. Through the development of eight data driven models including CatBoost, Ridge, XGBoost, LightGBM, MLPRegressor, HistGradient, Linear Regression and ExtraTrees. Ultimately, these models will produce meaningful insight that can assist healthcare professionals in prescribing to patients for more effective treatment plans. We measured the success of our models using R-squared, RMSE and MAE/MSE with the target variable being LN_IC50.

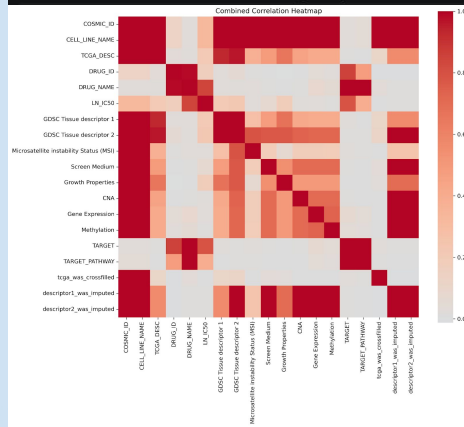
Data Acquisition

Source: All of our data was acquired from the Genomics of Drug Sensitivity in Cancer (GDSC) project, which collected data through large-scale screening of human cancer cell lines with various anti-cancer drugs.

Feature selection: Our original data was tabulated with 22 different columns, including the target variable, LN_IC50 (our measure of cancer drug sensitivity). Feature selection was done using baseline mutual-information analysis combined with repeated subsample-based selection. 9 of the 21 original features were selected for use during modeling.

Top 25 selected features by mutual information:

	Feature	MI_Score
0	DRUG_NAME_target_enc	0.643149
1	ploidy_wes	0.112369
2	ploidy_snp6	0.111351
3	mutational_burden	0.078163
4	Growth Properties_Suspension	0.028242
5	TARGET_PATHWAY_Mitosis	0.026173
6	Growth Properties_Adherent	0.021460
7	GDSC Tissue descriptor 1_Leukemia	0.017116
8	TARGET_PATHWAY_Metabolism	0.014283
9	GDSC Tissue descriptor 1_lymphoma	0.014073
10	TARGET_PATHWAY_Other	0.013964
11	TARGET_PATHWAY_Protein stability and degradation	0.013071
12	TARGET_PATHWAY_DNA replication	0.010219
13	TARGET_PATHWAY_Cell cycle	0.008713
14	TARGET_PATHWAY_WNT signaling	0.008364
15	TCGA_DESC_DLBC	0.007859
16	TARGET_PATHWAY_Unclassified	0.007817
17	TARGET_PATHWAY_Chromatin histone methylation	0.007365
18	TCGA_DESC_LAML	0.007087



Data Used for Modeling

DATASET	Description	Role
LN_IC50	Natural log of the half-maximal inhibitory concentration (IC50), used as the main drug-response target.	TARGET
DRUG_NAME	Name of the drug tested on the cell line.	Identifier
TCGA_DESC	Cancer type label based on The Cancer Genome Atlas abbreviation system.	Feature (IMP)
GDSC Tissue descriptor 1	Broad tissue or organ-system category for the cell line.	Feature (IMP)
GDSC Tissue descriptor 2	More specific tissue or disease subtype within the broader tissue category.	Feature (IMP)
Microsatellite instability Status (MSI)	Indicates whether the cell line shows microsatellite instability status, such as MSI or stable forms. The source paper explicitly includes MSI as one of the genomic factors analyzed in relation to drug response.	Feature
Growth Properties	Describes how the cell line grows in culture, such as adherent or suspension.	Feature
TARGET_PATH WAY	The biological pathway associated with the drug's target or mechanism of action.	Feature
mutational_burden	a biological feature describing how many mutations are present in the model/cell line.	Feature
ploidy_snp6	a ploidy estimate derived from SNP6 array data.	Feature
ploidy_wes	a ploidy estimate derived from whole-exome sequencing data.	Feature

- Added 3 more numerical genomic features for prediction; mutational_burden, ploidy_snp6, and ploidy_wes
- DRUG_NAME was encoded to the average LN_IC50 value of the drug
- Remaining categorical values were encoded using One-Hot Encoding
- Numerical values were standardized with StandardScaler.
- The rows marked UNCLASSIFIED under TCGA_DESC were included when running models with default parameters but dropped for the hyper tuned models because it adds additional unwanted noise for the prediction.
- DRUG_NAME cross validation target encoding was not done yet, but is part of the next phase.

Modeling

	Model	RMSE	MAE	R2
0	CatBoost	1.131118	0.844668	0.832297
1	XGBoost	1.145858	0.854324	0.827898
2	MLPRegressor	1.189713	0.893602	0.814473
3	HistGradientBoosting	1.225448	0.914313	0.803160
4	LightGBM	1.226727	0.915228	0.802749
5	ExtraTrees	1.249409	0.917899	0.795387
6	Ridge	1.344436	1.008067	0.763079

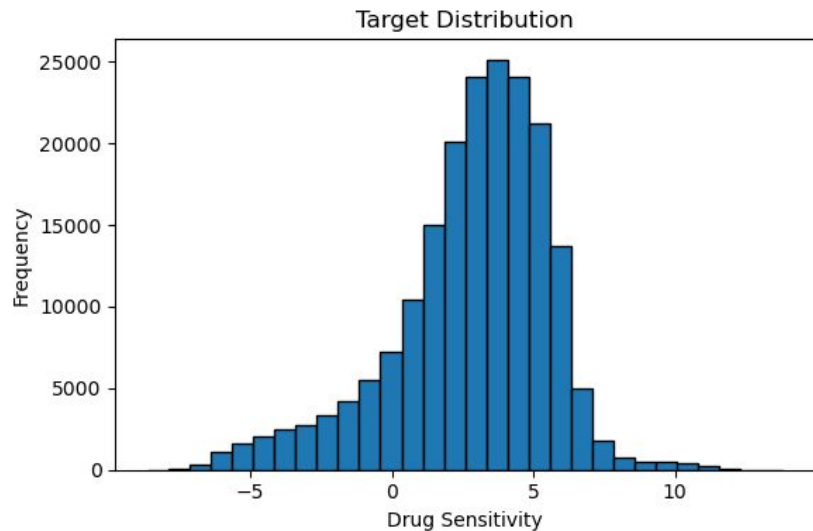
7 models ran on default parameters with the unclassified TCGA_DESC kept.

- Different models that ran the preprocessed data through default parameters as a way of understanding how different models react the new df.
- The hyper tuned parameter was done through 2 different ways. Some of the models were done using normal tuning by guessing and running multiple trials and the other method was by Optuna python library.
- Optuna is a Python library for automated hyperparameter tuning that intelligently searches for the best parameter combinations by iteratively testing trials and optimizing a chosen default/performance metric.

Catboost & Ridge

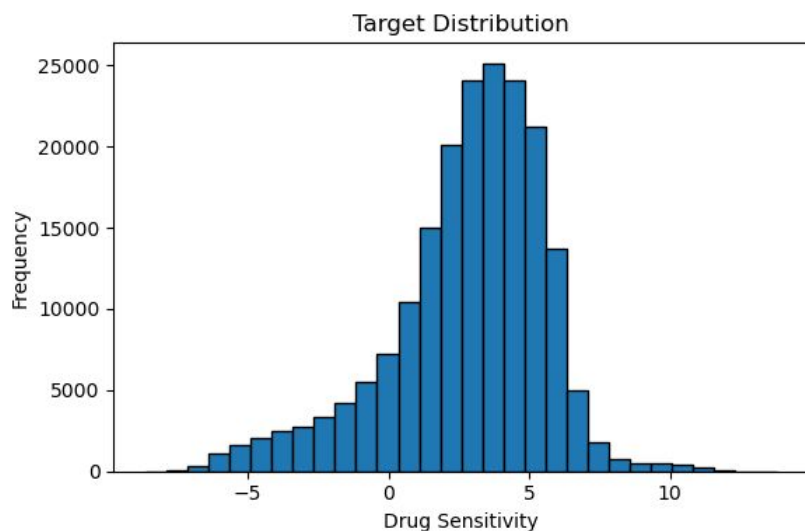
Ridge + Optuna

Best MSE: 1.7984836609418609
RMSe= 1.3410755612350338
Corresponding R^2 : 0.764268955980132
Corresponding MAE: 1.0059284319614514



Catboost

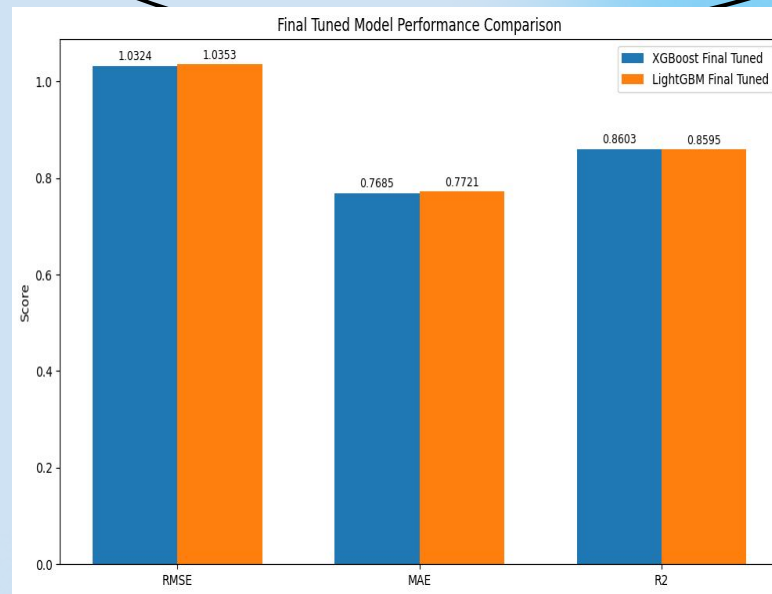
FINAL MODEL
RMSE: 1.1109239550772527
 R^2 : 0.8382321414640003
MAE: 0.8298048120373903



XGBoost & LightGBM

```
XGBRegressor(  
    objective='reg:squarederror',  
    random_state=42,  
    n_jobs=-1,  
    tree_method='hist',  
    n_estimators=2400,  
    learning_rate=0.058145405103894945,  
    max_depth=9,  
    min_child_weight=2,  
    subsample=0.8397243126557522,  
    colsample_bytree=0.6837348836421753,  
    gamma=0.3792103318134985,  
    reg_alpha=0.001137454986623023,  
    reg_lambda=5.184880698365349  
)
```

```
LGBMRegressor(  
    objective='regression',  
    random_state=42,  
    n_jobs=-1,  
    verbose=-1,  
    max_depth=12,  
    n_estimators=2000,  
    learning_rate=0.06459375058476324,  
    num_leaves=188,  
    min_child_samples=10,  
    subsample=0.7024966568890526,  
    colsample_bytree=0.7287148982717301,  
    reg_alpha=0.2213730715198598,  
    reg_lambda=0.657572365280675  
)
```

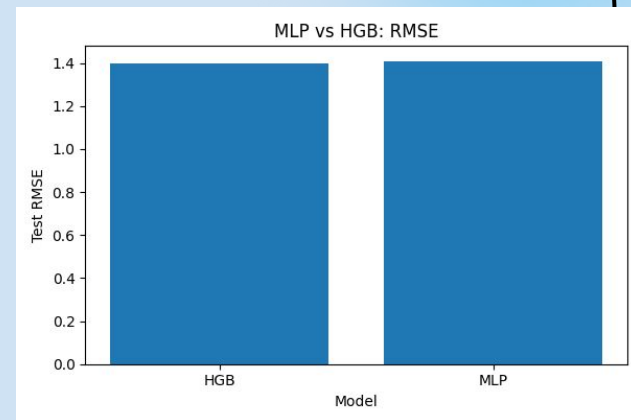
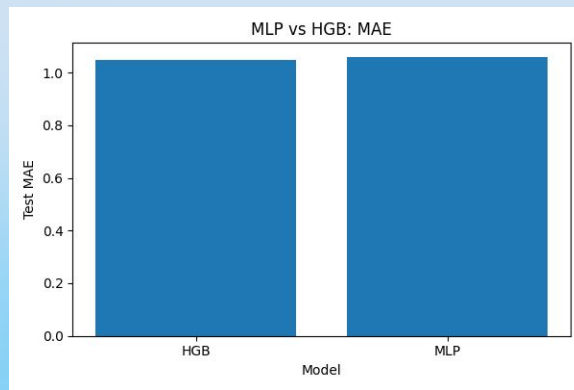
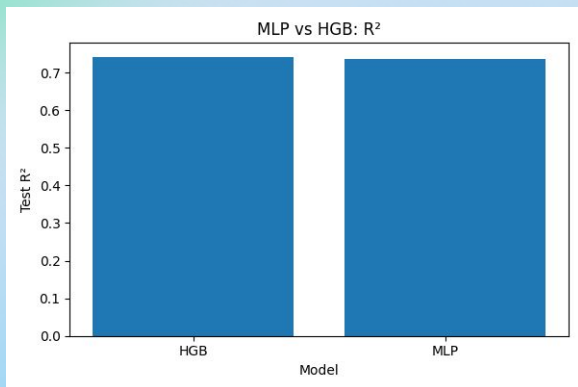


	Model	RMSE	MAE	R2
0	XGBoost Final Tuned	1.032373	0.768512	0.860300
1	LightGBM Final Tuned	1.035314	0.772103	0.859503

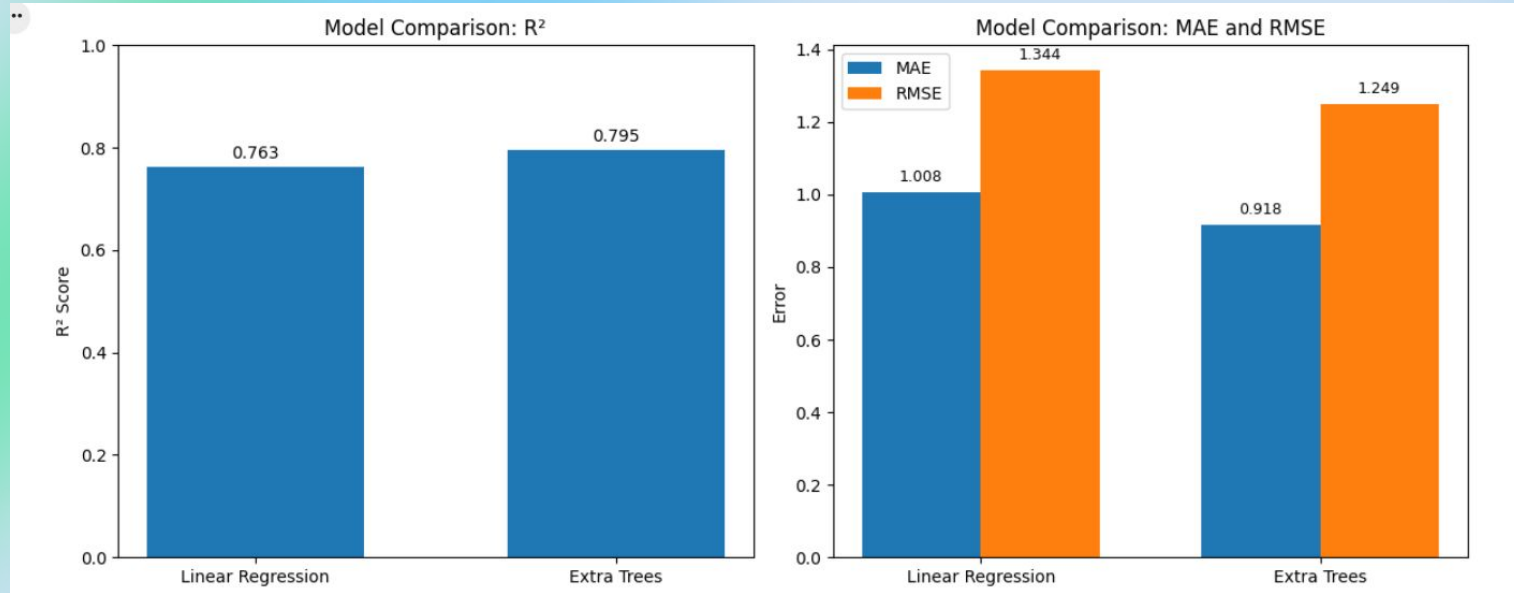
MLPRegressor & HistGradient

MLP vs HGB comparison:

Model	Test_RMSE	Test_MAE	Test_R2
HGB	1.396169	1.047938	0.741699
MLP	1.409077	1.059455	0.736900



Linear Regression & ExtraTrees



	Model	R^2	MAE	RMSE
0	Linear Regression	0.7631	1.0079	1.3443
1	Extra Trees	0.7954	0.9181	1.2494

Performance Evaluation

Model	RMSE	MAE	R2
XGBoost(w/ Optuna)	1.032373	0.768512	0.8603
LightGBM(w/ Optuna)	1.035314	0.772103	0.859503
Catboost(w/ Optuna)	1.050988	0.784145	0.855216
HistGradient(w/ Optuna)	1.099774	0.821224	0.841463
Catboost	1.110923	0.829804	0.838232
Extra Trees	1.2494	0.9181	0.7954
Ridge(w/ Optuna)	1.341	1.006	0.76426895
Linear Regression	1.3443	1.0079	0.7631
HistGradient	1.396169	1.047938	0.741699
MLPRegressor	1.409077	1.059455	0.7369

Deployment

- Current deployment idea: an interactive dashboard where users can input relevant biological/drug features and view predicted LN_IC50.
- Intended audience: healthcare professionals and researchers, so the output should be interpretable and easy to explore.
- Dashboard functions: predicted drug response, influential features, and comparisons across treatment options.

Link to Github Project and References

- 1) Link to Github Project:
<https://github.com/Temple-University-CIS-4496/cis-4496-project-cancer-pharmacogenomics/tree/master/Docs/Model>
- 2) Garnett, Mathew J., et al. "Systematic identification of genomic markers of drug sensitivity in cancer cells." Nature, vol. 483, no. 7391, Mar. 2012, pp. 570–575,
<https://doi.org/10.1038/nature11005>.
- 3) Woźniacki, Andrzej, et al. "A novel approach for predicting the survival of colorectal cancer patients using machine learning techniques and advanced parameter optimization methods." Cancers, vol. 16, no. 18, 20 Sept. 2024, p. 3205, <https://doi.org/10.3390/cancers16183205>.